

1. 8 QUEEN'S PROBLEM

Aim:

To develop a Python program that solves the 8-Queen's Problem using the Backtracking Algorithm, ensuring that no two queens attack each other on an 8×8 chessboard.

Objective :

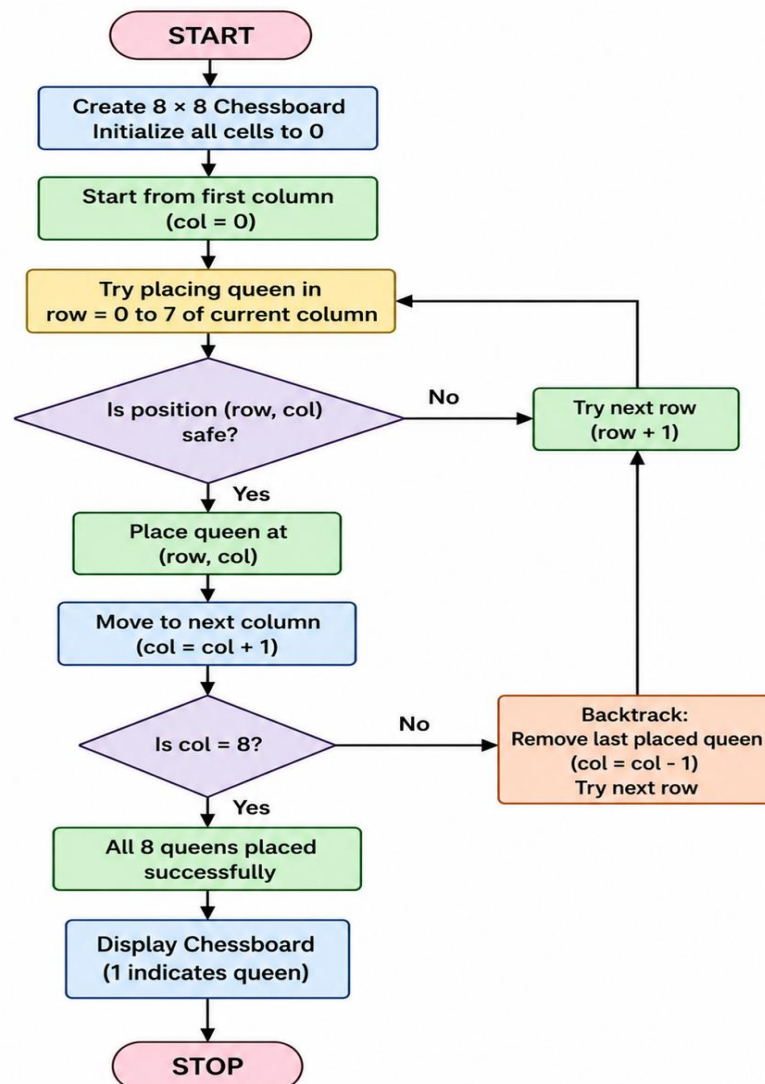
- To understand the concept of Backtracking.
- To place eight queens on an 8×8 chessboard.
- To ensure that no two queens share the same row, column, or diagonal.
- To generate one valid solution for the 8-Queen's problem.

Algorithm :

1. Start.
2. Create an empty 8×8 chessboard.
3. Begin placing queens from the first column.
4. For each row in the current column:
 - Check whether placing a queen is safe.
 - A position is safe if:
 - No queen exists in the same row.
 - No queen exists in the upper-left diagonal.
 - No queen exists in the lower-left diagonal.
5. If the position is safe:
 - Place the queen.
 - Recursively place the queen in the next column.
6. If all eight queens are placed:

- Display the solution.
- 7. If no safe position is available:
 - Remove the previously placed queen (Backtracking).
 - Try the next possible position.
- 8. Repeat until a solution is found.
- 9. Stop.

Flowchart :



Code:

```
N = 8
```

```
x = [-1] * N
```

```
def safe(row, col):
```

```
    for i in range(row):
```

```
        if x[i] == col or abs(x[i] - col) == abs(i - row):
```

```
            return False
```

```
    return True
```

```
def queen(row):
```

```
    if row == N:
```

```
        return True
```

```
    for col in range(N):
```

```
        if safe(row, col):
```

```
            x[row] = col
```

```
            if queen(row + 1):
```

```
                return True
```

```
    return False
```

```
if queen(0):
```

```
    print("Solution:")
```

```
    for i in range(N):
```

```
        for j in range(N):
```

```
            if x[i] == j:
```

```
                print(1, end=" ")
```

```
            else:
```

```
                print(0, end=" ")
```

```
    print()
```

else:

```
print("No solution")
```

Output:

```
Output
Solution:
1 0 0 0 0 0 0 0
0 0 0 0 1 0 0 0
0 0 0 0 0 0 0 1
0 0 0 0 0 1 0 0
0 0 1 0 0 0 0 0
0 0 0 0 0 0 1 0
0 1 0 0 0 0 0 0
0 0 0 1 0 0 0 0

=== Code Execution Successful ===
```

Result :

The Python program was successfully implemented using the Backtracking Algorithm to solve the 8-Queen's Problem. A valid arrangement of eight queens was obtained such that no two queens attack each other horizontally, vertically, or diagonally.